AI-AUGMENTED BACKEND DEVELOPMENT

AI BLOGNEST API

PHASE 2 – REQUIREMENTS ANALYSIS

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Submitted as a phase-wise project submission.

1. Introduction

The Requirements Analysis phase identifies what the AI BlogNest API must provide and the resources required to develop it. The requirements are divided into functional and non-functional requirements.

2. Functional Requirements

- User registration and login.
- Role-based access for Admin, Editor, Author and Reader.
- Create, read, update and delete blog posts.
- Manage categories, tags and comments.
- Generate blog content using AI.
- Summarize blog content using AI.
- Provide search and content retrieval through API endpoints.

3. Non-Functional Requirements

- Security for authentication and authorization.
- Reliable database operations and validation.
- Fast and consistent API responses.
- Maintainable and modular source code.
- Scalable architecture for future features.
- Proper error handling and logging.

4. Software Requirements

- Windows 10/11, macOS or Linux.
- Node.js and npm.
- Express.js.
- MongoDB with Mongoose.
- Visual Studio Code or similar code editor.
- Postman or Thunder Client for API testing.

5. Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent.
- RAM: 8 GB minimum; 16 GB recommended.
- Sufficient storage for the project, database and development tools.

6. Expected Outcome

A clear set of functional, non-functional, software and hardware requirements is prepared before implementation.

7. Conclusion

Requirements Analysis gives the development team a clear understanding of the system features, technologies and quality expectations.